Shane Patandin

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Education

Stony Brook University — Stony Brook, NY
Master of Science: Data Science — *Expected December 2026*

Stony Brook University — Stony Brook, NY
Bachelor of Science: Applied Mathematics and Statistics, Economics — GPA: 3.5 — *May 2025*

Relevant Coursework: Statistical Learning, Big Data Systems, Data Management, Data Mining, Machine Learning in Quantitative Finance, Mathematical Statistics, Econometrics

Experience

Artemis Speakeasy — Ronkonkoma, NY
Data Scientist Intern — *May 2026 – Present*

- Engineered a PostgreSQL data warehouse for 17 months of restaurant sales data (about $2.3M), with automated testing and a **16x faster refresh** (96s to 6s).
- Corrected an initial estimate for faster table turns **from $156K/yr to about $9K** by showing that revenue was constrained by demand rather than turnover, with 29% average occupancy.
- Quantified an annual decline of **about 14% in customer volume** after adjusting for seasonality, offset in part by an 8% rise in spend per guest driven by menu mix.

Stony Brook University — Stony Brook, NY
Research Assistant, Visual Analytics and Imaging Lab — *June 2026 – Present*

- Redesigned the capture pipeline so the compressed stream only locates and aligns the chart, then analyzes a **full resolution photo** from the glasses (1080x1440 vs 720x1280), with spoken cues guiding blind users to center it.
- Fixed an **out of memory crash** after about 5 captures by halving the frame buffer and plugging bitmap leaks, and added an OpenAI to Gemini to on device OCR fallback so API rate limits no longer crash analysis.
- Used real device data to remove an OCR gate rejecting 13 of 14 valid charts, and traced degraded detection across 1,000+ logged frames to the glasses SDK silently downscaling the stream on weak Bluetooth, affecting **92% of frames**.

Stony Brook University — Stony Brook, NY
Resident Assistant (RA) — *August 2024 – Present*

- Mentor and support 100+ residents as a first point of contact through on call duty, front desk check ins, and community programming for students from many countries.
- Surveyed 109 RAs with a partner and applied NLP to their written responses, finding that 58% said fewer events would improve quality and veteran RAs were **2.5x as likely** as first years to say requirements hurt quality (47% vs 19%); presented recommendations to the AICE committee.

Projects

Traffic Safety Analysis on Long Island | GitHub — April 2026

- Identified four accident archetypes across 33,000+ Long Island crash records using FAMD dimensionality reduction with k-Means, reaching a **silhouette score of 0.43 versus 0.16** for a k-Prototypes baseline.
- Built an XGBoost severity classifier on 26 engineered weather, infrastructure, and temporal features, with 5-fold CV confirming stable out of sample performance.
- Detected **12 accident corridors** along the LIE and Northern/Southern State Parkways with DBSCAN on projected coordinates, cross-referenced against the archetypes to target safety interventions.

Bank Term Deposit Prediction | GitHub — April 2026 – Present

- Tuned XGBoost, LightGBM, and Random Forest classifiers with Optuna on a highly imbalanced dataset, improving balanced accuracy over untuned baselines.
- Built a GPU-trained FT-Transformer in PyTorch that achieved the best balanced accuracy among non-leakage models, with SMOTE and scaling applied inside CV folds to prevent leakage.

Skills

Programming Languages: Python, R, SQL, SAS
Methods: Machine Learning, Regression, Classification, Clustering, NLP, Time Series, Statistical Modeling
Libraries: Scikit-learn, XGBoost, LightGBM, Optuna, PyTorch, OpenCV, Pandas, NumPy, Matplotlib, ggplot2
Databases & Tools: PostgreSQL, SQLAlchemy, Docker, Git, GitHub, Android Studio